



DEPARTMENT OF SOFTWARE ENGINEERING

Semester-V

Lab number : 01

**Lab Title: GitHub Identity, Student Portal Registration, and
Ubuntu Server Installation**

Name: Ayesha Afifa

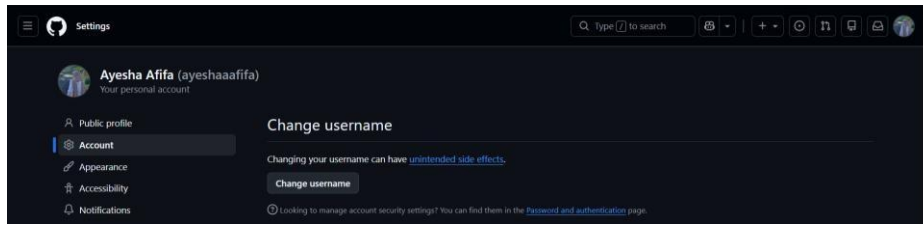
Registration number: FA24B1-SE-001

Course code: BSE-665

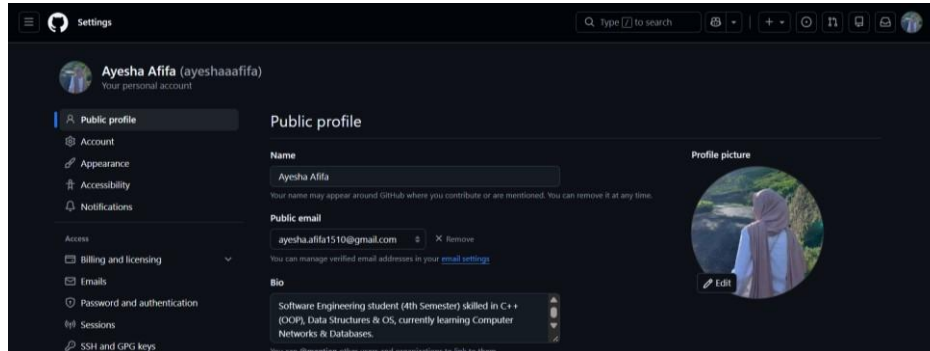
Section: A

GitHub username: ayeshaafifa

Task 1: Create and Configure Your GitHub Account



Renaming the GitHub account username in Account settings.



GitHub public profile updated with name, photo, and bio.

Task 2: Register in the Student Marks Portal



Authorizing Firebase GitHub Login to access the account for the Student Marks Portal.

Register:

A screenshot of the 'Complete registration' form. At the top, there is a black circle with the text 'ID'. Below this, the title 'Complete registration' is displayed. Under the title, there is a text prompt: 'Enter your registration number, course and section. The enrollment requires approval.' The form consists of three dropdown menus: 'Registration number' (with the value '2024-BSE-01'), 'Course' (with the value 'SE-665 — Cloud Computing'), and 'Section' (with the value 'Section A'). Below these dropdowns, there is a black button labeled 'Submit enrollment' and a light blue button labeled 'Sign out'.

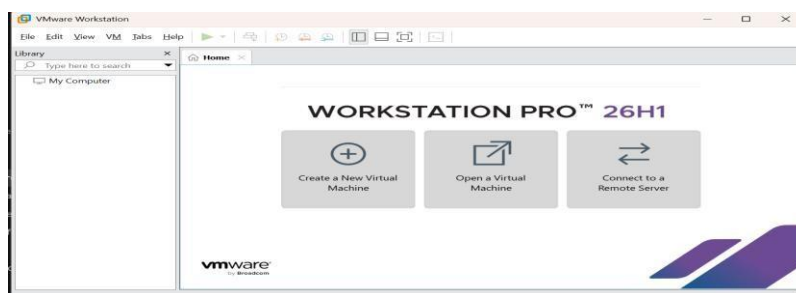
Completing enrollment with registration number, course, and section on the Student Marks Portal.

Enrollment:



Enrollment confirmed, status shown as pending administrator approval.

Task 3: Download VMware Workstation Pro and Ubuntu Server



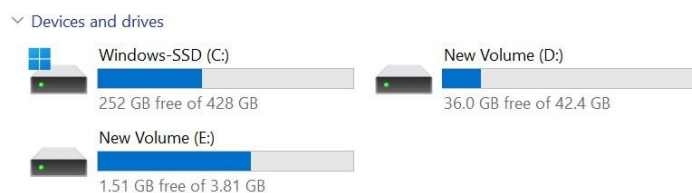
VMware Workstation Pro 26H1 home screen after installation.

Ubuntu iso file:



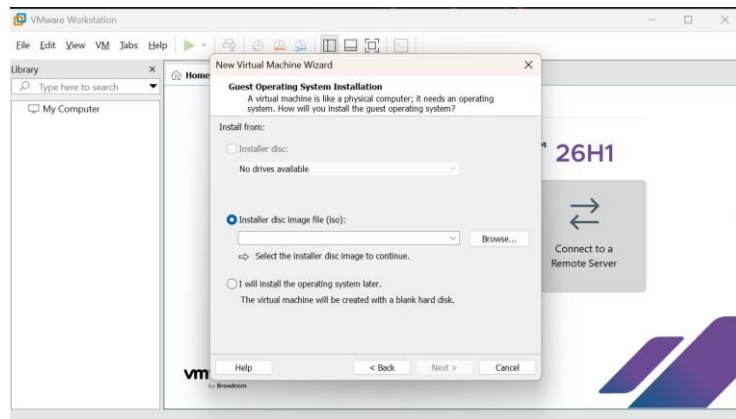
Download history confirming the Ubuntu Server ISO was downloaded.

Available Storage:



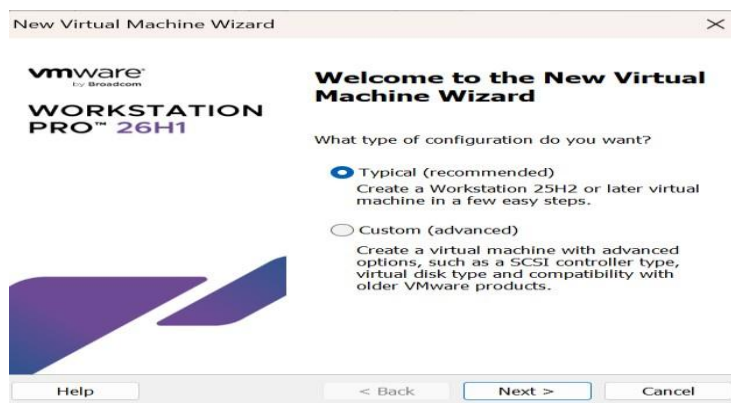
Available disk storage on the host machine before creating the VM.

Task 4: Create the Ubuntu Server Virtual Machine



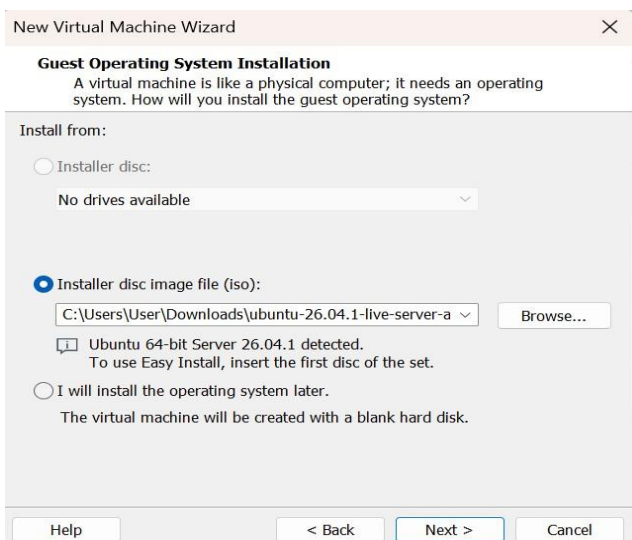
New VM Wizard set to install from an installer disc image file (ISO).

Select typical:



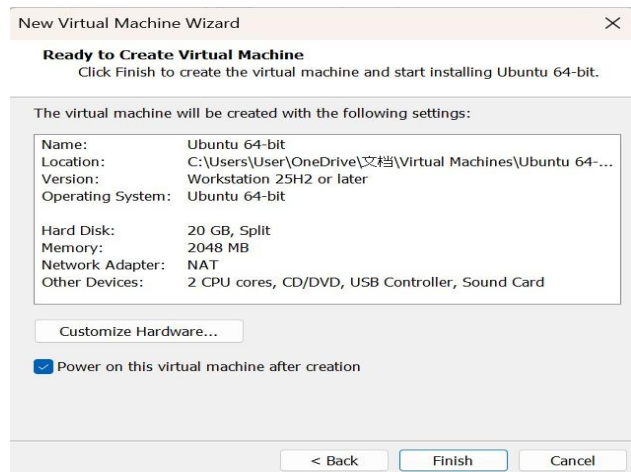
Typical (recommended) configuration selected for the new VM.

Browse ubuntu server file:



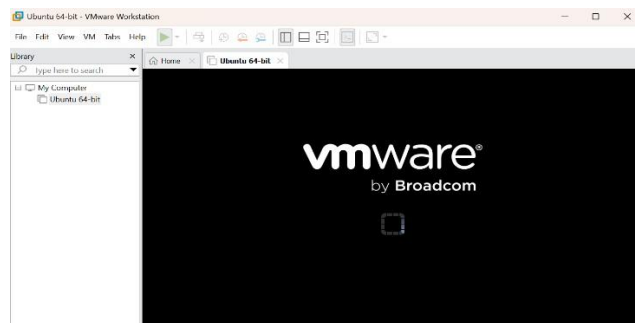
Ubuntu Server ISO selected as the install source; installer detects Ubuntu 64-bit Server 26.04.1.

Configuration Summary



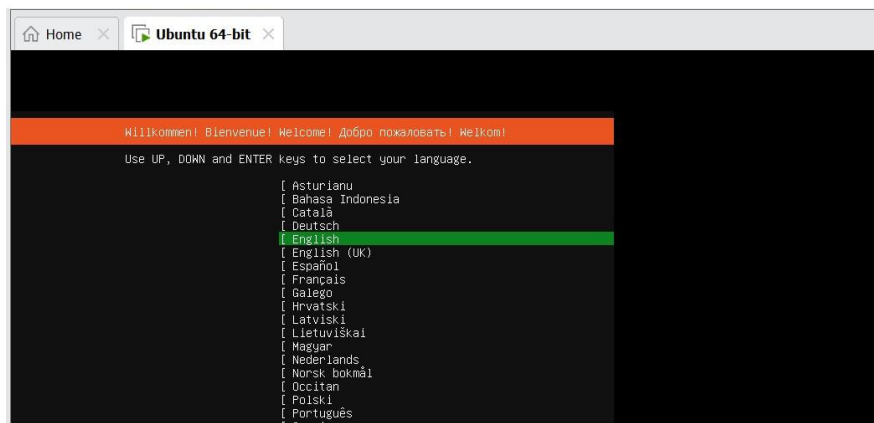
Final configuration summary before creation: name, 20 GB disk, memory, network.

Ubuntu installer boot waiting



VM powered on, waiting for the Ubuntu Server installer to boot.

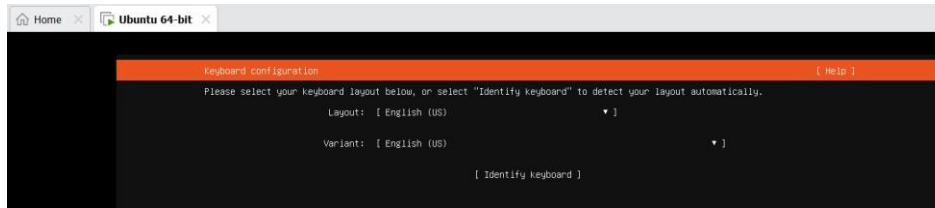
Task 5: Choose the Installation Language



Installer language selection screen with English highlighted.

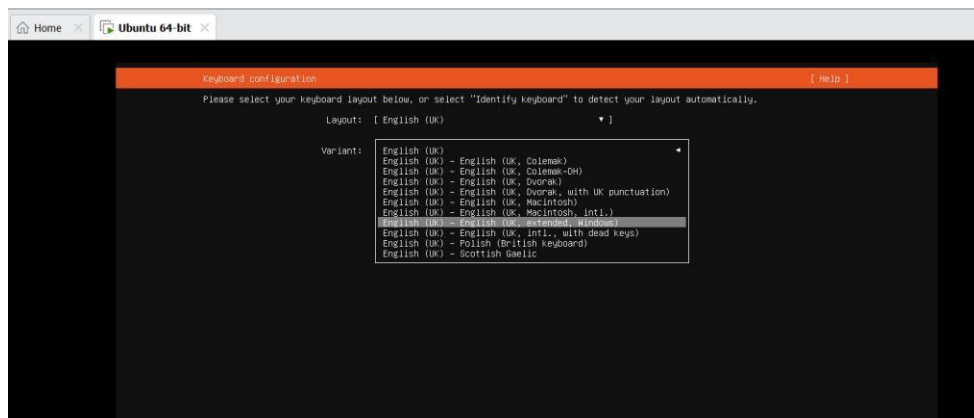
Task 6: Configure the Keyboard Layout

Layout:



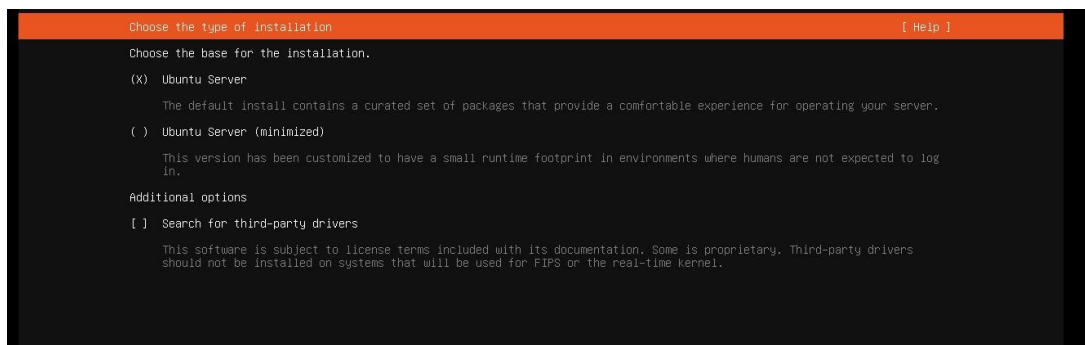
Keyboard layout screen showing English (US) selected.

Variant :



Keyboard variant dropdown expanded with English (UK) options.

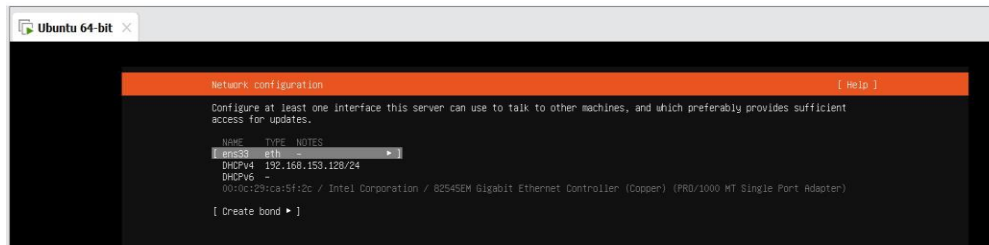
Task 7: Choose the Ubuntu Server Installation Type



Installation type screen with 'Ubuntu Server' selected.

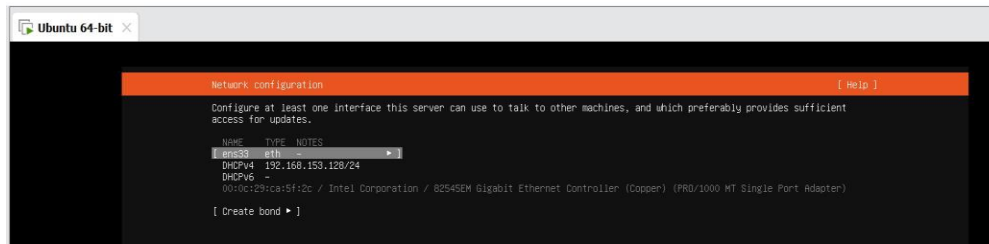
Task 8: Configure the Network

Network Interface:



Network configuration screen showing the detected interface (ens33).

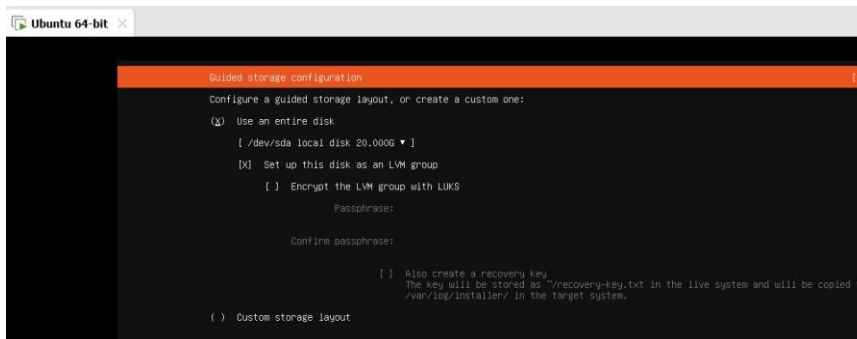
Installer network:



Interface confirmed to have received an IPv4 address via DHCP.

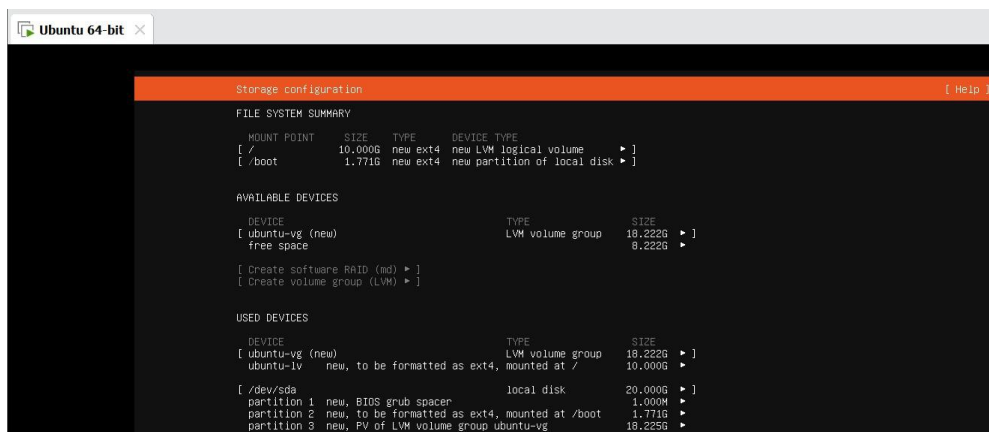
Task 9: Configure Storage

Guided Storage



Guided storage: 'Use an entire disk' selected on the 20 GB VMware disk.

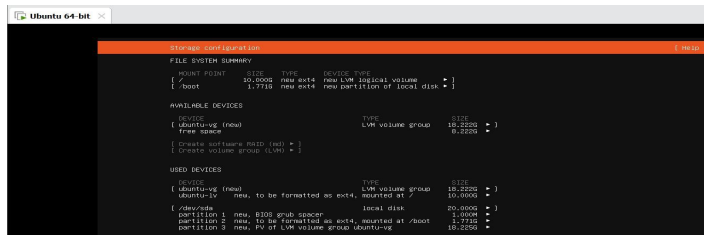
Storage Configuration:



Proposed file system layout (root and boot) on the virtual disk.

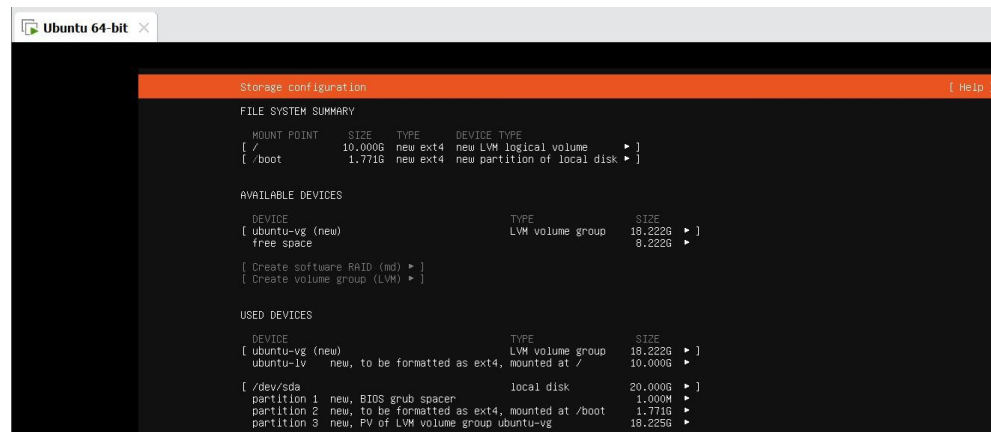
Task 10: Select the Installation Disk

Selected_disk



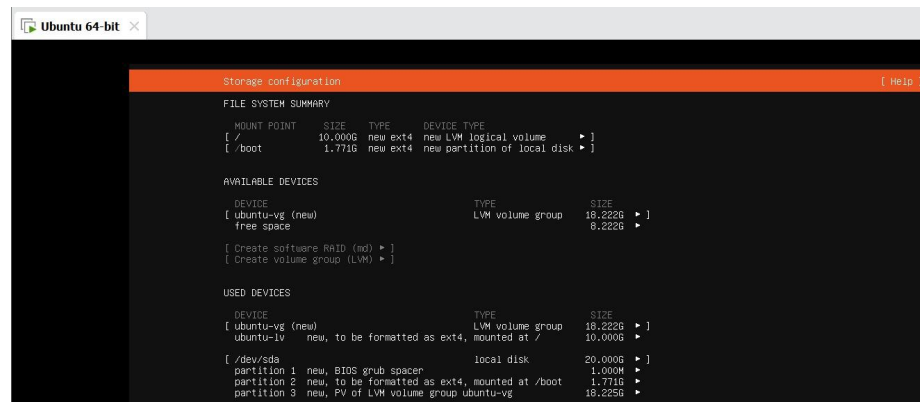
Selected installation disk confirmed as the VMware virtual disk.

Disk_details:



Disk details showing device name, partitions, and sizes.

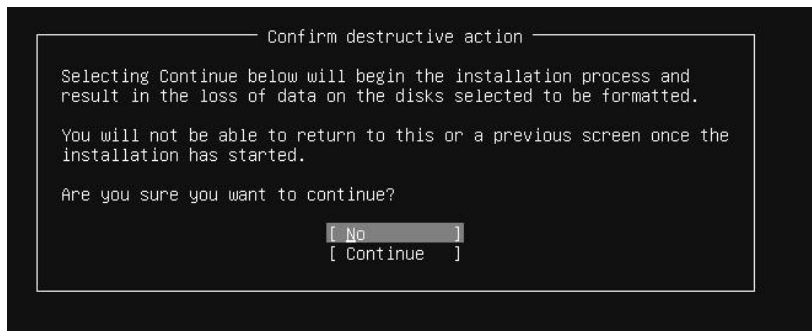
Task 11: Review the Partitions



Complete partition layout reviewed before confirming changes.

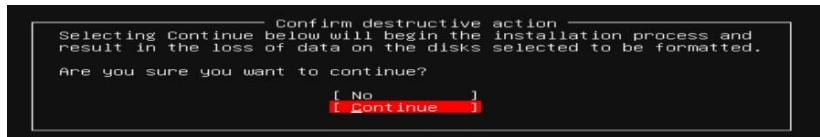
Task 12: Confirm the Storage Changes

Storage Warning



Destructive-action warning shown, with 'No' highlighted by default.

Storage Confirmed:



Selection moved to 'Continue' to confirm the storage changes.

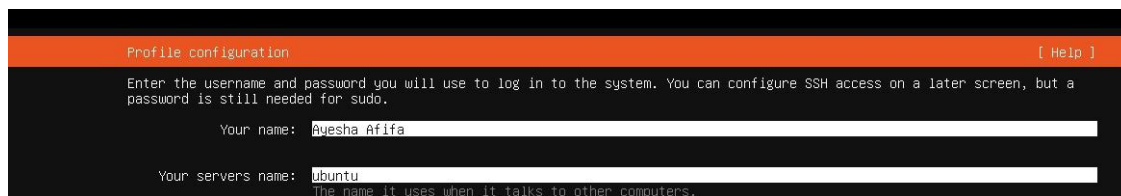
Task 13: Configure the Ubuntu Profile

Actual Name:



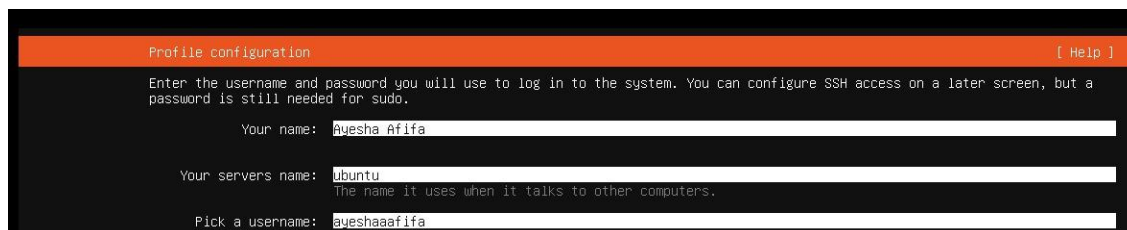
Profile configuration: actual name 'Ayesha Afifa' entered.

Server name:



Server name set to 'ubuntu' during profile configuration.

Username:



Login username 'ayeshaaafifa' chosen during profile configuration.

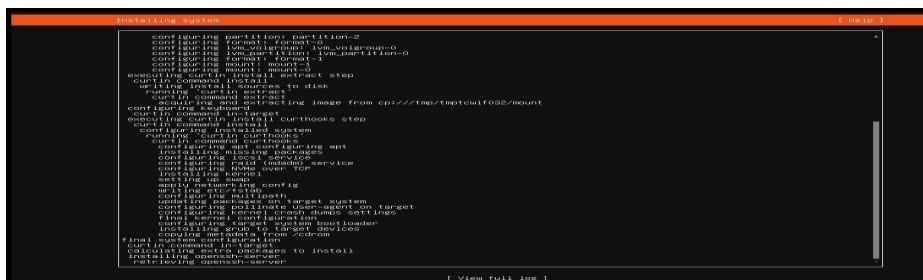
Review profile:



The screenshot shows the 'Profile configuration' window with a title bar containing '[Help]'. The main text reads: 'Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.' Below this are four input fields: 'Your name:' with 'ayesha Afifa', 'Your servers name:' with 'ubuntu', 'Pick a username:' with 'ayeshaafifa', and 'Choose a password:' with '*****'. A 'Confirm your password:' field is also present with '*****'.

Completed profile with name, server name, username, and password set.

Task 14: Complete the Software Installation

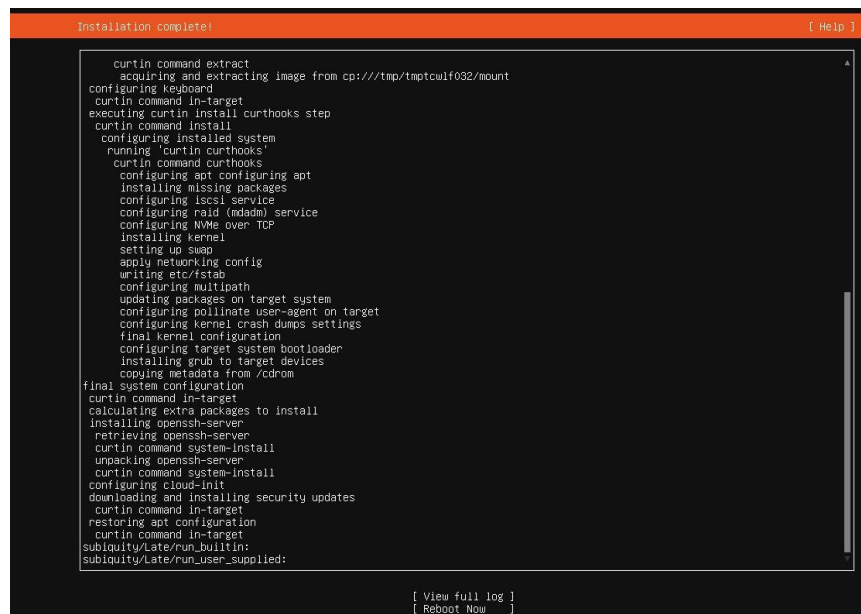


The screenshot shows a terminal window titled 'Installing system' with a '[Help]' button. It displays a list of tasks being performed by the installer, including: configuring partitions, configuring keyboard, configuring network, configuring apt, installing missing packages, configuring iSCSI service, configuring RAID (mdadm) service, configuring NVMe over TCP, installing kernel, setting up swap, applying networking config, writing etc/fstab, configuring multipath, updating packages on target system, configuring pollinate user-agent on target, configuring kernel crash dumps settings, final kernel configuration, configuring target system bootloader, installing grub to target devices, copying metadata from /cdrom, final system configuration, curtin command in-target, calculating extra packages to install, installing openssh-server, retrieving openssh-server, curtin command system-install, unpacking openssh-server, curtin command system-install, configuring cloud-init, downloading and installing security updates, curtin command in-target, restoring apt configuration, curtin command in-target, subiquity/late/run_builtins, and subiquity/late/run_user_supplied. At the bottom, there are links for '[View full log]' and '[Reboot Now]'.

Software installation in progress, showing the installer's package log.

Task 15: Reboot and Verify the Ubuntu Identity

Installation complete:



The screenshot shows the 'Installation complete!' window with a '[Help]' button. It displays a list of tasks that have been completed, including: curtin command extract, acquiring and extracting image from cp:///tmp/tmptcwl032/mount, configuring keyboard, curtin command in-target, executing curtin install curthooks step, curtin command install, configuring installed system, running 'curtin curthooks', curtin command curthooks, configuring apt, configuring apt, installing missing packages, configuring iSCSI service, configuring RAID (mdadm) service, configuring NVMe over TCP, installing kernel, setting up swap, applying networking config, writing etc/fstab, configuring multipath, updating packages on target system, configuring pollinate user-agent on target, configuring kernel crash dumps settings, final kernel configuration, configuring target system bootloader, installing grub to target devices, copying metadata from /cdrom, final system configuration, curtin command in-target, calculating extra packages to install, installing openssh-server, retrieving openssh-server, curtin command system-install, unpacking openssh-server, curtin command system-install, configuring cloud-init, downloading and installing security updates, curtin command in-target, restoring apt configuration, curtin command in-target, subiquity/late/run_builtins, and subiquity/late/run_user_supplied. At the bottom, there are links for '[View full log]' and '[Reboot Now]'.

Installation complete, ready to reboot into Ubuntu Server.

Ubuntu login screen:



The screenshot shows the Ubuntu login screen with the text: 'ubuntu login:', 'ubuntu login: ayeshaafifa', and 'Password: _'.

Ubuntu login prompt after reboot with username 'ayeshaaaafifa'.

Ubuntu terminal login:

```
ubuntu login:
ubuntu login: ayeshaafifa
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 13 09:41:30 AM UTC 2026

System load:  0.0      Processes:      216
Usage of /:   50.5% of 9.75GB   Users logged in: 0
Memory usage: 21%      IPv4 address for ens33: 192.168.153.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ayeshaaaafifa@ubuntu:~$
```

Terminal login successful, showing system info and IPv4 for ens33.

Ubuntu identity verified:

```
ayeshaaaafifa@ubuntu:~$ _
```

Terminal prompt confirming the logged-in Ubuntu identity.

Task 16: Find the Ubuntu Server IP Address

Ip addr command

```
ayeshaaaafifa@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:ca:5f:2c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29ca5f2c
    inet 192.168.153.129/24 metric 100 brd 192.168.153.255 scope global dynamic ens33
        valid_lft 970sec preferred_lft 970sec
    inet6 fe80::20c:29ff:fe5f:2c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaaaafifa@ubuntu:~$
```

Output of 'ip addr' showing network interface details for ens33.

Ubuntu ip address

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:ca:5f:2c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29ca5f2c
    inet 192.168.153.129/24 metric 100 brd 192.168.153.255 scope global dynamic ens33
        valid_lft 970sec preferred_lft 970sec
    inet6 fe80::20c:29ff:fe5f:2c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaaaafifa@ubuntu:~$ _
```

Close-up of the assigned IPv4 address 192.168.153.129.

Task 17: Connect to Ubuntu Server from Windows

Windows ssh command

```
Command Prompt - ssh ayesi  X + v
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh ayeshaafifa@192.168.153.129
The authenticity of host '192.168.153.129 (192.168.153.129)' can't be established.
ED25519 key fingerprint is SHA256:aUM/s3XiaoR5wggnc8CTR97T/BERhtjbwQj4T88Nee8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

SSH connection started from Windows CMD; host key fingerprint prompt shown.

Ssh_fingerprint:

```
ayeshaafifa@ubuntu: ~ X + v
The authenticity of host '192.168.153.129 (192.168.153.129)' can't be established.
ED25519 key fingerprint is SHA256:aUM/s3XiaoR5wggnc8CTR97T/BERhtjbwQj4T88Nee8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.153.129' (ED25519) to the list of known hosts.
ayeshaafifa@192.168.153.129's password:
```

Fingerprint accepted ('yes'); password prompt shown for SSH login.

ssh login:

```
ayeshaafifa@192.168.153.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 05:49:29 PM UTC 2026

System load:  0.03               Processes:    223
Usage of /:   50.6% of 9.75GB    Users logged in:  0
Memory usage: 23%               IPv4 address for ens33: 192.168.153.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ayeshaafifa@ubuntu:~$ |
```

Successful SSH login from Windows into the Ubuntu Server VM.

ssh identity:

```
Last login: Sun Sep 13 17:51:21 2026 from 192.168.153.1
ayeshaafifa@ubuntu:~$ |
```

Terminal prompt confirming the Ubuntu Server identity over SSH.
